# *Analysis and Design of Algorithms*

NUS National University of Singapore

Algorithms

CS3230

**Tutorial**

Week 11

# Question 1

In the graph coloring problem, you are given an undirected graph $G = (V, E)$ and asked to color the vertex with a set of colors such that the colors of $v_1$ and $v_2$ must be different if $(v_1, v_2) \in E$. In the optimization version of the problem, the task is to find the minimum number of colors required to color the graph. In the decision version, the question is whether there is a coloring of the graph using $k$ or fewer colors. Which of the following are true?

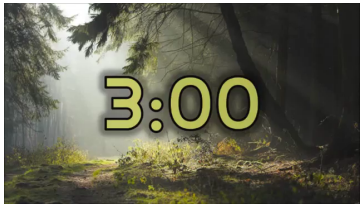
(I) If we can solve the optimization problem for graph coloring in polynomial time, we would be able to solve the decision problem for graph coloring in polynomial time.

(II) If we can solve the decision problem for graph coloring in polynomial time, we would be able to solve the optimization problem for graph coloring in polynomial time.

(III) If the decision problem for graph coloring cannot be solved in polynomial time, the optimization problem for graph coloring cannot be solved in polynomial time.

(IV) If the optimization problem for graph coloring cannot be solved in polynomial time, the decision problem for graph coloring cannot be solved in polynomial time.

- (I) only
- (II) only
- (I) and (III) only
- (II) and (IV) only
- All true.



# Question 1

**Answer:** E All true

**(I)** Solve opt → solve decision:
Find min colors, then check whether min ≤ $k$

**(II)** Solve decision → solve opt:
Sequentially solve for number of colors to find min (or binary search)

**(III)** Contrapositive of (I)

**(IV)** Contrapositive of (II)

# Question 2

TSP: Given a complete weighted directed graph $G = (V, E)$ and a threshold $k$, does there exist a tour (cycle) that visits all vertices in $V$ exactly once with cost at most $k$?

Give an instance of the problem $G = (V, E)$ and a value $k = 100$, We discover that the lowest cost tour has cost 53.

Which of the following are **valid** certificates that show that the answer to the TSP problem is YES?

(I) A tour that visits all vertices in $V$ exactly once with cost 53.

(II) A tour that visits all vertices in $V$ exactly once with cost 152.

(III) A tour that visits all vertices in $V$ exactly once with cost 70.

- (I) only
- (II) only
- (III)
- (I) and (III)

# Question 2

**Answer:** D (I) and (III)

Valid certificate can be used to **prove** the answer is YES in polynomial time.

- **(I)** Can. Check that all vertices are visited once, first and last vertices are the same, consecutive pairs of vertices in tour are in edge set, total edge weight of 53 ≤ 100. All can be done in poly time
- **(II)** Cannot. Does not show answer is YES as cost 153 > 100
- **(III)** Can. Like (I), cost 70 is still ≤ 100.

# Question 3

Which of the following imply P=NP?

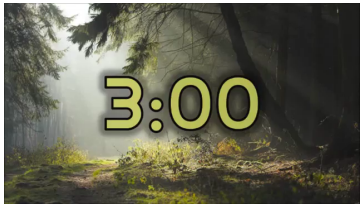
(I) There is a problem in P that is also in NP-COMPLETE.

(II) There is a problem in P that is also in NP.

(III) There is a problem in P that is also in co-NP.

(IV) There is a problem in NP that is also in NP-HARD.

- (I) only
- (I) and (II) only
- (II) and (III)
- (I) and (IV)



# Question 3

**Answer:** A

Which implies P = NP?

- (I) There is a problem in **P** and **NP-COMPLETE**
  **True**: if there is an NP-HARD problem that can be solved in poly time, all problems in NP can be solved in poly time, so P = NP.

- (II) There is a problem in **P** and **NP**.
  **False**: Any problem in P is also in NP. So this does not imply P = NP

# Question 3

**Answer:** A

Which implies P = NP?

- (III) There is a problem in **P** and **co-NP**.
  **False**: Any problem in P is also in co-NP. So this does not imply P = NP

- (IV) There is a problem in **NP** and **NP-HARD**.
  **False**: We already know NP-COMPLETE problems are in both NP and NP-HARD, so this does not imply P = NP.

# Question 4

$L$ is an NP problem. One day, Mr Oh proves that it is impossible to find an algorithm to solve $L$ in polynomial time. After this big event, which of the following classes of problems are now **known** to be **not solvable** in polynomial time?

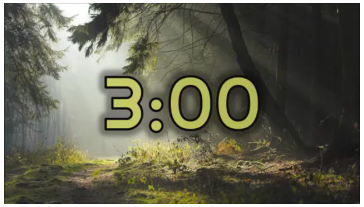
(I) All NP-HARD problems.

(II) All NP problems

(III) All NP-COMPLETE problems.

(IV) FACTOR

- (I) only
- (II) only
- (III) only
- (I) and (III)
- (I), (II), and (III)
- (I), (III), and (IV)
- (I), (II), (III), and (IV)



# Question 4

**Answer:** D (I) and (III)
Suppose a problem *L* in NP is proved to be not solvable in poly time, which are **known** to be **not solvable** in poly time:

- (I) All NP-HARD problems:
  **True**, otherwise will be able to solve *L*
- (II) All NP problems:
  **False**, P ⊆ NP can still be solved in poly time

# Question 4

**Answer:** D (I) and (III)
Suppose a problem *L* in NP is proved to be not solvable in poly time:

- (III) All NP-COMPLETE problems:
  **True**: Same as (I)
- (IV) Factor:
  **Not known (False)**, as Factor is not known to be NP-HARD.